A High-Performance

AD831-Based Mixer

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I. Project Proposal

Moving beyond basic filter designs, this project focuses on the heart of many communication systems: the frequency mixer. I intend to build a high-performance mixer using the Analog Devices AD831 monolithic mixer IC. The primary goal is to create a robust and demonstratable module that not only confirms the theoretical principle of frequency conversion (e.g., RF to IF) but also allows for hands-on investigation of two critical practical aspects in RF design: **disturbance** and **impedance matching**. The final product will be a **self-made PCB** with SMA connectors, enabling precise measurements and fully prepared video / onsite demonstration.

II. Circuit Schematic

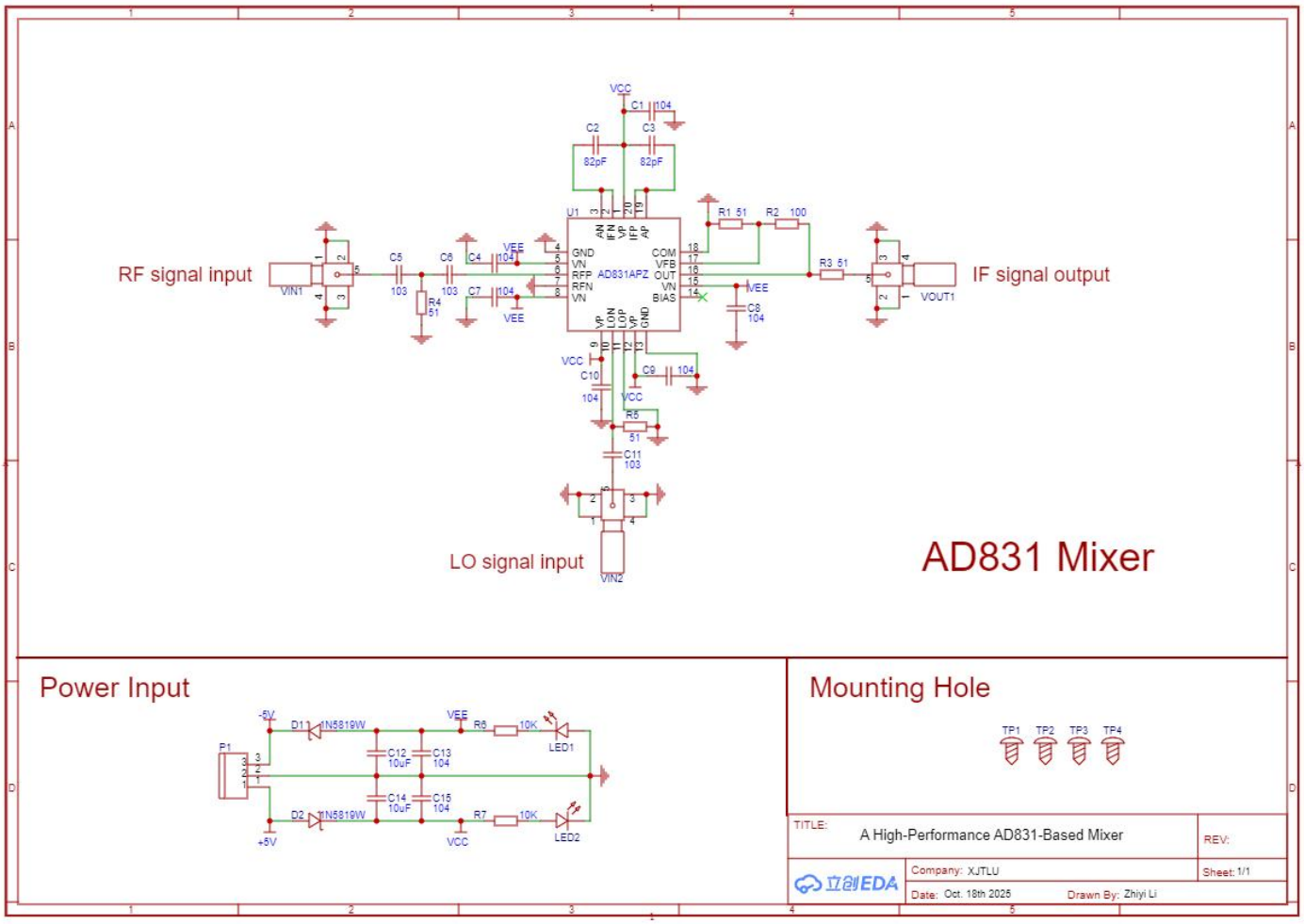


Figure 1. Circuit Schematic

III. Component List (BOM)

Category	Quantity	Value	Supplier	Footprint	Unit Price
Capacitor	8	100 nF	LCSC	C0603	0.0147
Capacitor	2	82 pF	LCSC	C0603	0.0197
Capacitor	3	10 nF	LCSC	C0603	0.0137
Capacitor	2	10 uF	LCSC	C0603	0.2032
Diode	2	1N5819W	LCSC	SOD-123F	0.0478
Connector	3	SMA	LCSC	CONN-SMD	2.81
LED	1	Blue	LCSC	LED0603	0.0227
LED	1	Red	LCSC	LED0603	0.0634
Connector	1	5.08-3P	LCSC	CONN-TH	1.223
Resistor	4	51 Ω	LCSC	R0603	0.0071
Resistor	1	100 Ω	LCSC	R0603	0.0066
Resistor	2	10k Ω	LCSC	R0603	0.0067
Copper Pillar	4	M3	Taobao	-	0.42
IC	1	AD831APZ	Taobao	PLCC-20	8.8

List 1. Bill of Materials

Component overall cost: 21 RMB
PCB + Stencil cost: 2 + 20 = 22 RMB
Soldering cost: 5 RMB

List 2. Other Cost

Overall cost: 48 RMB per Product.

IV. Reference

[1]. ANALOG DEVICES (ADI) AD831 official datasheet

[2]. National Undergraduate Electronic Design Contest Preparation Guide and Case Analysis – Based on LCEDA (《全国大学生电子设计竞赛备赛指南与案例分析——基于立创 EDA》)